

Youssef Elebiary

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Summary

Computer science student at Cairo University with a strong foundation in AI, machine learning, and full-stack development. Experienced in building real-world applications using Python, C++, Flask, and modern ML frameworks. Certified by IBM, Google, and Huawei in AI and cybersecurity. Passionate about developing intelligent systems and seeking an internship or entry-level role in AI or software development to apply skills, learn, and contribute to impactful projects.

Experience

NLP Engineer Intern, Elevvo Pathways, Cairo Jul 2025 – Aug 2025

- Fine-tuned a **BART-base** model for abstractive text summarization, achieving a **ROUGE-1 score of 39.9** on the CNN-DailyMail benchmark.
- Built end-to-end classification pipelines for **Sentiment Analysis** and **Fake News Detection** using TF-IDF and optimized standard classifiers.
- Implemented unsupervised topic modeling using **LDA** and **NMF**, achieving a superior **coherence score of 0.77** with NMF to extract latent themes from unstructured news data.

Information Technology Intern, FPi-Future Pharmaceutical industries, Cairo Aug 2025 – Sep 2025

- Enhanced a custom task management system by adding a "comment" feature that improved teamwork.
- Contributed to an Order Receiving mobile app (Kotlin + Django API), partially integrated with the JD Edwards (JDE) ERP system.
- Applied back-end development best practices while gaining hands-on experience with .NET and Angular.

Education

BSc in Computer Science, **Cairo University** Sep 2023 – Jul 2027

- GPA: 3.6

Certifications

Microsoft Machine Learning Engineer, DEPI Dec 2025

IBM AI Engineering Specialization, IBM – Coursera Sep 2024

HCIA – Artificial Intelligence, Huawei ICT Academy Sep 2024

Object Oriented Programming Specialization, University of London – Coursera Sep 2024

Google AI Essentials, Google – Coursera Sep 2024

Projects

RescuNet - AI Disaster Response Platform github.com/YoussefElebiary/RescuNet

- Developed a **Graph Neural Network (GINEConv)** to predict road traversability on highly imbalanced, sparse graphs and engineered a **C++ routing engine** to solve VRP in milliseconds.
- Integrated **YOLOv11** models for RGB hazard/human detection and Thermal human detection, alongside an **NLP LSTM** model to classify and prioritize distress signals.
- Built a full **React 19** website with **MapLibre** and **FastAPI** enabling users to visualize live drone feeds, **perform real-time text analysis**, and generate dynamic rescue routes.

AI Image Captioning

github.com/YoussefElebiary/VisionScribe

- Developed a captioning architecture using **InceptionV3** for feature extraction and **Bidirectional LSTMs** for sequence generation.
- Achieved a **BLEU-1 score of 0.54** and **BLEU-4 of 0.15**, validating the model's ability to generate semantically accurate descriptions.
- Implemented **Beam Search** decoding to improve caption quality and deployed the system via a **Flask** REST API with a web interface.

Fake News Classifier

github.com/YoussefElebiary/TruthGuard

- Engineered a dual-model classification system using **XGBoost** (Standard) and **LightGBM** (Lite), achieving a consistent **99.8% F1-Score** and **99.9% Precision**.
- Developed a **Flask** web interface that allows users to toggle between high-precision and lightweight inference modes for real-time analysis.

Image Editing Software

github.com/YoussefElebiary/PhotoWizard

- Built a desktop GUI app in **C++** using **Qt** to apply visual filters to images with real-time previews.
- Supported multiple image formats and optimized performance for smooth user interaction.

Recipe Finder Website

github.com/YoussefElebiary/RecipeHub

- Designed a cooking platform for users to browse, search, and filter recipes using **Django**.
- Built the backend with **Django** and styled the frontend with custom **HTML/CSS**.

Skills

Technical Skills

Python, C++, HTML/CSS
Scikit-learn, PyTorch
Django, FastAPI
Git

Personal Skills

Problem Solving
Communication
Teamwork
Time Management

Languages

Arabic – Native
English – C1
French – A2
German – A2

Extracurricular Activities

MSP Tech Club - Helwan University

Head of AI & Data Science

Jan 2026 – Present

- Leading the AI committee to design technical workshops and mentor junior members.
- Orchestrating the season plan to align with current industry trends in Data Science.

AI Committee Member

May 2025 – Dec 2025

- Collaborated on internal AI projects and organized community tech initiatives.